

Eigenvalue Problem:

Eigenvalue problem concern the solution of the system

$$AX = \lambda X \quad \text{————— (1)}$$

where A is a given square matrix and vector X and scalar λ are unknown,

Clearly $X=0$ is a solution of (1). But this is of no interest, and we want to find solution vector $X \neq 0$ of (1), called eigenvectors of A .

We can find eigenvectors only for certain values of the scalar λ ; these values λ for which an eigenvector exists are called the eigenvalues of A .

Note: • eigenvalues are also called characteristic values or latent roots and eigenvectors are also called characteristic vectors.

- Revise properties of eigenvalues from first semester.

Power method: In many engineering problems, it is required to compute the numerically largest eigenvalue and the corresponding eigenvector. In such cases, we use the ^{iterative} power method. The procedure will be clear by the following questions:

(Here we find dominant eigenvalue λ such that $|\lambda|$ is greater than the absolute values of the other eigenvalues) [^{* Here we repeat the process till $x^{(n)} - x^{(n-1)}$ becomes negligible}]

Que Find the largest eigen value and the corresponding eigen vector of the matrix $\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$ using power method. Take $[1, 0, 0]^T$ as initial eigen vector.

Sol Let the initial approximation to the required eigen vector be $X = [1, 0, 0]^T$. Then

$$AX = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ -0.5 \\ 0 \end{bmatrix} = \lambda^{(1)} X^{(1)}$$

↓
Take numerically largest value outside

So, the first approximation to the eigenvalue is 2 and the corresponding eigen vector $X^{(1)} = [1, -0.5, 0]^T$

$$\text{Now } AX^{(1)} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -0.5 \\ 0 \end{bmatrix} = \begin{bmatrix} 2.5 \\ -2 \\ 0.5 \end{bmatrix} = 2.5 \begin{bmatrix} 1 \\ -0.8 \\ 0.2 \end{bmatrix} = \lambda^{(2)} X^{(2)}$$

$$AX^{(2)} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -0.8 \\ 0.2 \end{bmatrix} = \begin{bmatrix} 2.8 \\ -2.8 \\ 1.2 \end{bmatrix} = 2.8 \begin{bmatrix} 1 \\ -1 \\ 0.43 \end{bmatrix} = \lambda^{(3)} X^{(3)}$$

$$AX^{(3)} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 0.43 \end{bmatrix} = \begin{bmatrix} 3 \\ -3.43 \\ 1.86 \end{bmatrix} = -3.43 \begin{bmatrix} -0.87 \\ +1 \\ -0.54 \end{bmatrix} = \lambda^{(4)} X^{(4)}$$

$$AX^{(4)} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} -0.87 \\ +1 \\ -0.54 \end{bmatrix} = \begin{bmatrix} -2.74 \\ +3.41 \\ -2.08 \end{bmatrix} = 3.41 \begin{bmatrix} -0.80 \\ +1 \\ -0.61 \end{bmatrix} = \lambda^{(5)} X^{(5)}$$

Dr. Nishu Gupta

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$$AX^{(5)} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} -0.80 \\ 1 \\ -0.61 \end{bmatrix} = \begin{bmatrix} -2.6 \\ 3.41 \\ -2.22 \end{bmatrix} = 3.41 \begin{bmatrix} -0.76 \\ 1 \\ -0.65 \end{bmatrix} = \lambda^{(6)} X^{(6)}$$

Clearly $\lambda^{(5)} = \lambda^{(6)} = 3.41$ and $X^{(5)} = X^{(6)} \approx \begin{bmatrix} -0.8 \\ 1 \\ -0.6 \end{bmatrix}$

Hence the largest eigenvalue is 3.41 and the corresponding eigenvector is $[-0.8, 1, -0.6]^T$.

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